

SOFTWARE REQUIREMENTS SPECIFICATION

Diet and Nutrition System

Document Type: Software Requirements Specification (SRS)

System: Diet and Nutrition Management System

Version: 1.0

1. Introduction

1.1 Purpose

The purpose of the Diet and Nutrition System is to help users manage their diet, nutrition, meal plans, food intake, and personal nutrition goals. The system provides food information, personalized meal plans, progress tracking, and nutrition recommendations.

1.2 Scope

The system allows users to register, maintain profiles, set nutrition goals, view meal plans, log food intake, track progress, and generate reports. Nutritionists can create diet plans and provide recommendations. Administrators can manage users, food information, and reports.

1.3 Definitions

SRS – Software Requirements Specification. User – person who uses the application to manage diet and nutrition. Nutritionist – professional who provides diet recommendations. Admin – administrator who manages system data.

2. Overall Description

2.1 Product Perspective

The system is a web/mobile application consisting of a user interface, application/business logic, nutrition services, meal plan services, report services, user management, data access, and a database.

2.2 User Classes

3. Functional Requirements

FR1 – Registration

The system shall allow a new user to create an account using required personal details.

FR2 – Login and Authentication

The system shall authenticate users using valid login credentials and reject invalid credentials.

FR3 – Profile Management

Users shall be able to view and update age, gender, weight, height, and relevant health information.

FR4 – Nutrition Goals

Users shall be able to set nutrition goals such as weight loss, weight gain, or maintenance.

FR5 – Food Information

The system shall provide food items and their nutrition information such as calories, protein, fat, and carbohydrates.

FR6 – Meal Plan

The system shall generate and display personalized meal plans based on user information and nutrition goals.

FR7 – Food Intake Logging

Users shall be able to record daily meals and food intake.

FR8 – Nutrition Calculation

The system shall calculate calorie and nutrient intake from recorded food information.

FR9 – Progress Tracking

The system shall track progress such as weight, calories, nutrients, and goal progress.

FR10 – Reports

The system shall generate nutrition and progress reports for users.

FR11 – Nutritionist Recommendations

Nutritionists shall be able to review user information and provide diet recommendations or adjust plans.

FR12 – Admin Management

Administrators shall be able to manage users, food database information, and system reports.

4. Non-Functional Requirements

4.1 Performance

The system should respond to normal user requests within a reasonable time and support multiple users.

4.2 Security

User credentials and personal information shall be protected. Access shall be controlled according to user roles.

4.3 Usability

The interface should be simple, clear, and easy to use on supported web/mobile devices.

4.4 Reliability

The system should maintain accurate records and minimize data loss.

4.5 Maintainability

The system should use modular components so that individual services can be updated easily.

4.6 Scalability

The system should be capable of supporting additional users, food items, and services in the future.

5. External Interface Requirements

5.1 User Interface

The application shall provide screens for registration, login, profile, nutrition goals, food search, meal plans, food logging, progress, and reports.

5.2 Software Interfaces

The system may communicate with a nutrition/food database API and a database management system.

5.3 Hardware Interface

The application shall work on standard smartphones, tablets, laptops, or desktop computers with supported browsers or the supported mobile application.

5.4 Communication Interface

Web-based communication shall use secure HTTP/HTTPS connections where applicable.

6. System Constraints

6.1

The quality of recommendations depends on the accuracy of user-provided information and food database data.

6.2

The system requires appropriate database and network availability for online functions.

6.3

The system is intended as a nutrition-management application and does not replace professional medical diagnosis or treatment.

7. Data Requirements

User Data

User ID, name, email, password credentials, phone number, and profile information.

Food Data

Food ID, food name, category, calories, protein, fat, and carbohydrates.

Meal Data

Meal log ID, date, meal type, food intake, calories consumed, and notes.

Goal Data

Goal type, target value, current value, start date, and end date.

Report Data

Report ID, user ID, report type, generated date, and report content.

8. System Modules

8.1 User Module

Registration, login, profile management, goals, food logging, and progress tracking.

8.2 Nutrition Module

Nutrition calculation, food information, and nutrient analysis.

8.3 Meal Plan Module

Personalized meal plan generation and plan updates.

8.4 Nutritionist Module

User review, recommendations, and diet-plan adjustment.

8.5 Report Module

Progress analysis and nutrition report generation.

8.6 Admin Module

User management, food database management, and report management.

9. UML Diagrams Included

9.1

Use Case Diagram – shows interactions between User, Nutritionist, Admin, and the system.

9.2

Class Diagram – shows system classes, attributes, methods, and relationships.

9.3

Activity Diagram – shows the workflow of the diet and nutrition system.

9.4

Sequence Diagram – shows the sequence of interactions for generating a personalized diet plan.

9.5

State Diagram – shows the lifecycle of a user's diet plan.

9.6

Component Diagram – shows the major software components and their dependencies.

10. Conclusion

10.1

The Diet and Nutrition System provides a structured way to manage personal nutrition information, meal plans, food intake, and progress. The system also supports nutritionist recommendations and administrative management, making it suitable as a diet and nutrition management software project.

Appendix A – User Roles Summary

User Type	Responsibilities
User	Manage profile, nutrition goals, meal plans, food intake, progress, and reports.
Nutritionist	Review user information, create diet plans, and provide recommendations.
Admin	Manage users, food database, and reports.